

# Ratio Reasoning: Convert Measurements Between Systems

Lesson 3-7

**Name:** Type your name

**Date:** Today's date

**Class:** Class period

---

## My Notes

Level 2 Standard



### TODAY'S OBJECTIVES

**Content Objective:** I can use ratio reasoning to convert measurements between the customary and metric systems.

**Language Objective:** I can explain a conversion between systems using the words conversion factor, approximately, customary, and metric.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



**Fill in each blank as we go. Use the Word Bank to help you.**



### WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Conversion factor

Customary system

Metric system

Approximately

Ratio table



Tap any word to see its meaning and a picture.

1

The ratio that trades one unit for another is a

 .

2

Inches, pounds and gallons belong to the

system.

3

Centimeters, grams and liters belong to the

system, built on tens.

4

Close to but not exactly equal is

— written with  $\approx$ .

5

A table whose every column holds the same comparison is a

 .

## Write About the Math

**Is 100 kilometers per hour faster or slower than 100 miles per hour, and how do you know?**

*¿100 kilómetros por hora es más rápido o más lento que 100 millas por hora, y cómo lo sabes?*

Say your answer to a partner first. Then write it, using at least two of these words:

☐

**Conversion factor**

*Factor de conversión*

☐

**Customary system**

*Sistema usual*

☐

**Metric system**

*Sistema métrico*

☐

**Approximately**

*Aproximadamente*

☐

**Ratio table**

*Tabla de razones*

**Model:** A mile is longer than a kilometer ( $1 \text{ km} \approx 0.6 \text{ mi}$ ), so 5 miles is the longer race even though the numbers match. — Students reason about UNIT SIZE before touching arithmetic — same number, different units, different lengths.

### Start

Complete the frames. Use the word bank.

*Completa las oraciones. Usa el banco de palabras.*

- The conversion factor is  $1 \text{ km} \approx \underline{\hspace{1cm}}$  mi. Scaling that up, 100 kilometers  $\approx \underline{\hspace{1cm}}$  miles, so the sign means about  $\underline{\hspace{1cm}}$  miles per hour. That is  $\underline{\hspace{1cm}}$  than 100 miles per hour, because in one hour you would travel  $\underline{\hspace{1cm}}$  distance.
- My answer is  $\underline{\hspace{1cm}}$  because  $\underline{\hspace{1cm}}$ .

*Mi respuesta es  $\underline{\hspace{1cm}}$  porque  $\underline{\hspace{1cm}}$ .*

### Build

Write two connected sentences. Use because, so, or but.

*Escribe dos oraciones conectadas. Usa porque, entonces o pero.*

- My claim is  $\underline{\hspace{1cm}}$ .
- My evidence is  $\underline{\hspace{1cm}}$ , so  $\underline{\hspace{1cm}}$ .

*Mi evidencia es  $\underline{\hspace{1cm}}$ , entonces  $\underline{\hspace{1cm}}$ .*

---

---

## Explain

Write a claim, give math evidence, and explain your reasoning.

*Escribe una afirmación, da evidencia matemática y explica tu razonamiento.*

---

---

---

---

---

---

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.